ELSEVIER

Earth and Planetary Science Letters 136 (1995) 461–473

EPSL

Paleomagnetism and magnetostratigraphy of the traps from Western Taimyr (northern Siberia) and the Permo-Triassic crisis

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Received 16 May 1995; revised 15 September 1995; accepted 18 September 1995

Abstract

The Siberian traps were emplaced near the Permo-Triassic boundary and are among the largest continental igneous provinces. A 1750 m thick section of traps from the western Taimyr (72.9°N, 84°E) has been sampled. This section is equivalent to the lower and middle part of the total Siberian traps section. Thirty-one oriented hand samples were studied and demagnetised by alternating fields and heating cycles. A stable characteristic component passes the fold test and is thought to be primary. The mean direction is $D = 84°$, $I = 75°$ with $\alpha_{95} = 10°$ ($N = 29$) and the corresponding pole is 59°N, 150°E. The bottom part (660 m) is of reversed polarity, the middle part (900 m) is normal and the uppermost level (150 m) is mixed. Correlations with reference Permo-Triassic magnetostratigraphic timescales are discussed. The most probable correlation indicates an age quite close to the Permian/Triassic boundary and a duration for the volcanism of less than 1 m.y. The data are therefore consistent with the hypothesis that this volcanism triggered the Permian/Triassic crisis.

1. Introduction

Since 1980, when Alvarez et al. proposed that the Cretaceous/Tertiary (K/T) crisis had an extraterrestrial origin, a long debate has occurred between the supporters and opponents of this explanation [1–9]. An alternative to this are drastic climate modifications due to the Deccan traps volcanism. Similarly, it was predicted, first by Morgan [10] and then by others [11–15], that the Permo-Triassic (P/Tr) biological crisis was caused by the huge Siberian traps volcanism. A critical test of this hypothesis would be to obtain the exact age and duration of this volcanism: if the volcanism is older or younger than the P/Tr boundary, it could not be the cause of the biological crisis. Long-lasting volcanism will have a lesser impact on the climate than short, violent volcanism. Paleomagnetism and magnetostratigraphy may place constraints on the age and duration of this volcanism.

2. The Siberian traps

The Siberian traps are widespread in the Tunguska syncline of the Siberian Platform and on Gorny

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SSDI 0012-821X(95)00183-2

Taimyr (Fig. 1). These traps consist of hypabyssal intrusions and effusive and explosive facies of mafic magmatic rocks. They cover a surface of at least $1.5 \cdot 10^6$ km^2 and the cumulative thickness is more than 3 km, reaching as much as 4 km in some places.

The volcanism may be divided into five cycles, which are themselves subdivided, on geochemical grounds, into suites [16] (see Table 5). Cycles I and II have a more explosive character than subsequent cycles and are not present everywhere. Cycles III

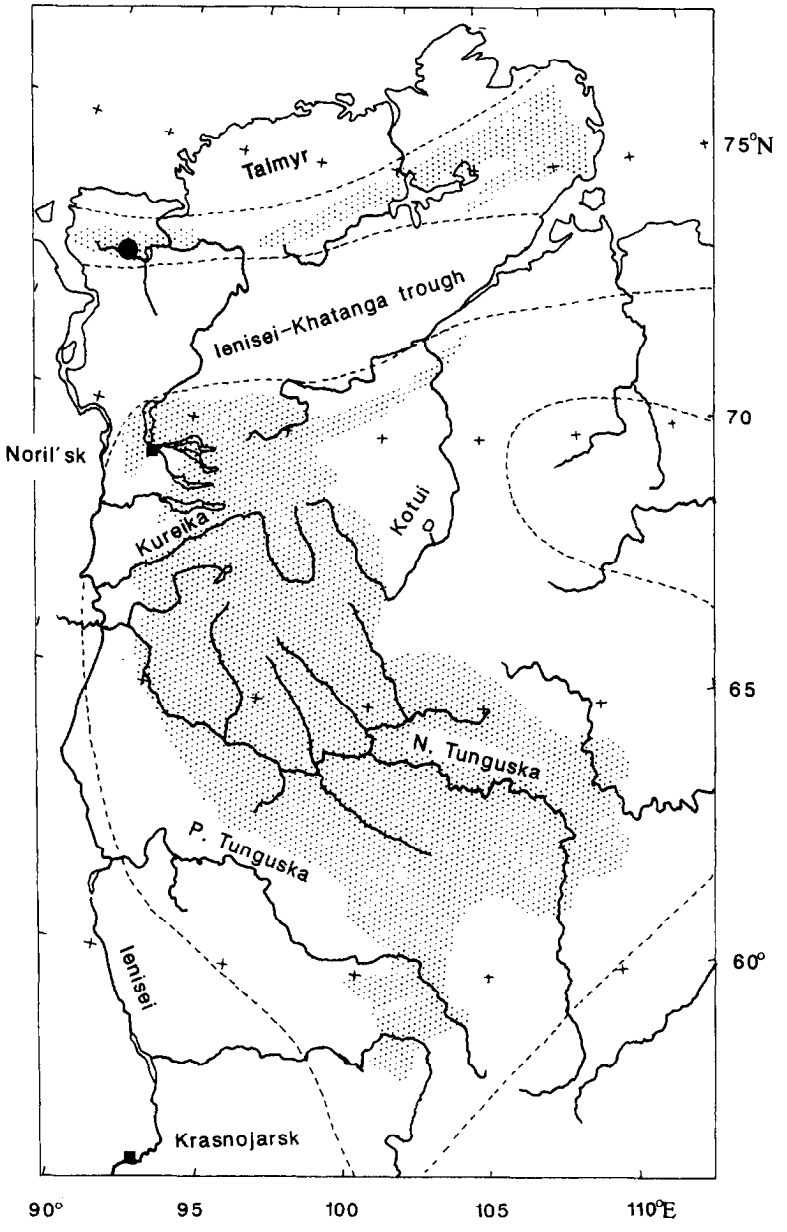


Fig. 1. Geological outline of northern Siberia. Stipple = Permo-Triassic effusives; broken line = limit of the intrusives; ● = location of the Syradasay River section.

and IV are widespread. Cycle V is found only in a restricted area. More precise and refined petrological and geochemical studies have been carried out since the work of Zolotukhin and Al,Mukhamedov [16], mostly in the Noril'sk area [17–19], but they have not changed our overall view of the Siberian traps volcanism.

2.1. Age of the volcanism

The Siberian traps unconformably overlie older Paleozoic and Proterozoic rocks; the youngest rocks covered by the traps are Upper Permian sediments. Within the traps themselves, sedimentary levels are rare and only local. These levels contain Upper Permian and Lower Triassic floras. Cycles I and II are considered as Upper Permian in age and the subsequent ones as Lower to Middle Triassic [16].

Precise isotopic ages of the Siberian flood basalts fall in a narrow range. Baksi and Farrar [20] analysed samples from the Tunguska Basin (Morongovsky and Mokulaevky suites) by $^{39}Ar/^{40}Ar$ using incremental step heating and obtained ages of around 244 Ma. Dalrymple et al. [21,22] studied the age spectrum of biotites and plagioclases from lavas and intrusions in the Noril'sk and Talnakh districts by Ar/Ar using total and step fusion and they obtained an age of 249 ± 2 Ma for the Noril'sk I and Talnakh ore intrusions. Slightly younger ages (2%) obtained on basalts were due to argon loss and alteration.

Renne and Basu [11] analysed three different samples from the bottom, middle and upper parts (Ivakinsky, Gudchinsky and Nerakarsky suites) of the traps in the Noril'sk and Putorana regions by $^{39}Ar/^{40}Ar$ and by laser heating on whole rocks and on plagioclase separates. These workers obtained closely coincident ages with a mean of 248.5 ± 2.4 Ma and interpreted them as indicating a short duration of less than 1 m.y. for the volcanism and an age close to the P/Tr boundary.

Campbell et al. [12] used U/Pb on zircons from an intrusion that was comagmatic with the traps from the Noril'sk region and obtained an age of 248.0 ± 3.7 Ma. These authors argued strongly for involvement of the eruption in the P/Tr crisis.

To conclude, even the most reliable age determinations were not able to resolve the time duration of the volcanism. The apparent age, which is about 250 Ma, is close to the P/Tr boundary as given by Claoué-Long et al. [23] at 251.2 ± 3.4 Ma and by Gradstein et al. [24] at 248.2 ± 4.8 Ma.

3. The Syradasay River section in western Taimyr

The Taimyr Peninsula is at present separated from the Siberian Platform by the E–W trending Ienisei–Khatanga Trough, although the traps found on this peninsula do belong to the same sequence as those from the Siberian Platform more to the south and can be easily correlated with the platform traps. The main trap field in Taimyr occurs in the southern Byranga mountain range, where it is folded and broken by various faults. The volcanic rocks in western Taimyr are represented by basalts which fill E–W trending basins. Ranges composed of terrigenous deposits bearing coal of Upper Paleozoic age also occur in western Taimyr.

The best section of western Taimyr is situated in the middle Syradasay River, on the left bank on the northern flank of a horst-anticline (72.9°N, 84°E). The regional dip of the lava varies from block to block, from northwest to northeast. Dips change from 70–80° at the bottom of the section to practically horizontal at the top. The entire thickness is about 1.7 km (Fig. 2).

The section is subdivided on geochemical grounds, with, from bottom to top, three suites being recognised: Syradasaysky, Labaksky and Verhnetamsky. The lavas of the Syradasaysky suite belong to the alkali-basalt series and are represented by trachybasalts (with Ti-augite), subalkaline basalts with a higher content of olivine, and hawaiites. The silica content of the lower Syradasaysky 'undersuite' varies between 44.5 and 48%, whereas that of the hawaiites of the upper undersuite varies between 49.5 and to 52%. These alkali-basalts differ from the overlying tholeiitic basalts by virtue of their 4–6% total content of alkaline elements and owing to the appearance of pink Ti-augites. The overlying Labaksky suite is mainly composed of tholeiitic basalts of a more homogeneous composition. Subdivisions within these suites are based on finer petrological variations and on the appearance of a higher content of olivine within the Labaksky suite.

The lowermost Syradasaysky is correlated with

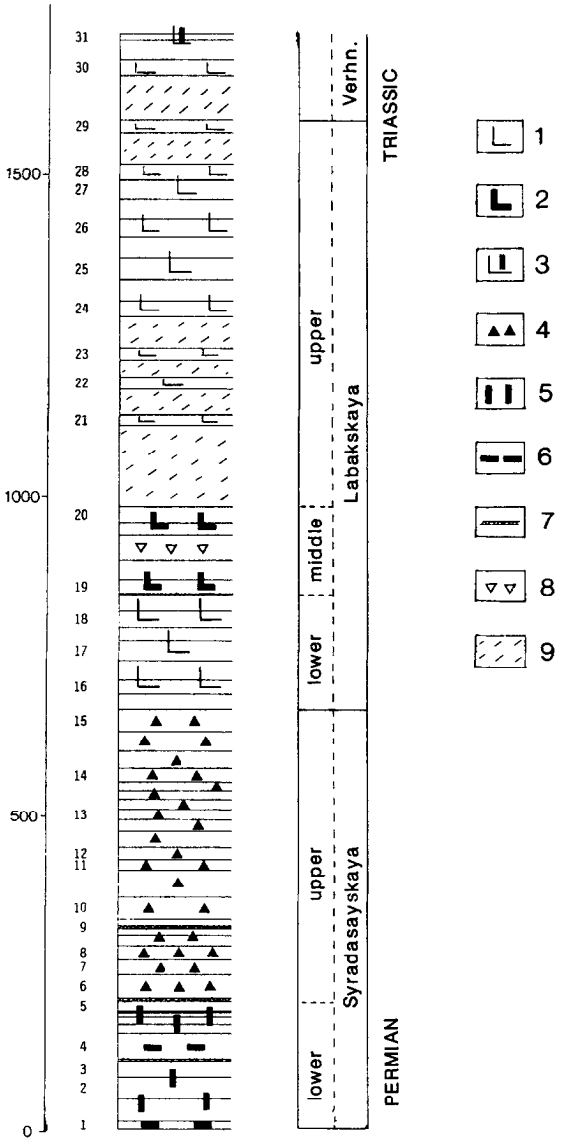


Fig. 2. The Syradasay River section. *1* = aphyric tholeiites; *2* = tholeiites with more olivine; *3* = tholeiites with olivine and plagioclase; *4* = alkali basalts and hawaiites; *5* = subalkaline basalts; *6* = trachybasalts with Ti-augites; *7* = volcaniclastics; *8* = tuffs; *9* = gaps in outcrops. Thicknesses are in metres.

the Ambarninsky and the Ivakinsky suites in the Noril'sk region, where the latter represent Cycle I. The Labaksky suite corresponds to Cycles II and III. The uppermost suite, Verhnetamsky, is similar to the Morongovsky suite in the Noril'sk region, which forms a part of Cycle IV. The Syradasaysky section therefore represents the first two-thirds of the total Siberian trap volcanism. Sedimentary intercalations are rare and very thin. It is important to note at this juncture that this section was folded after the volcanism and before the Jurassic, which is usually horizontal. This allows us to conduct a fold test, which is usually impossible on the Siberian Platform itself where all the lavas are horizontal.

3.1. Age of the Syradasay River section

The P/Tr age of this section may be determined from floral remnants in two levels. The lower of these is situated at the bottom of the section in adjacent areas among coal-bearing terrigenous deposits. The floral remnants are represented by, among others, *Phyllotheca minaeana* Radcz., *Paracalamites communis* Gorel., *Cordaites insignis* Radcz., and *Lepeophyllum ex gr. actaeonelloides* (Gein.) Radcz. (unpublished determinations by M.V. Verbitskaya, VSEGEI, St. Petersburg). These remnants are characteristic of the Upper Permian Tatarian stage of the Angarida paleocontinent. The upper level is situated at the bottom of the Verhnetamsky suite, and the remnants here—again among others—are *Cladophlebis angustipinnula* Mogutch., *Sphenopteris kirjamkensis* Pryn., and *Bernoullia, Tersiella jarakcensis* Schved. (unpublished determinations by M.V. Turante and I.A. Dobruskina, RAN, GIN, Moscow). These are typical of the Korvunchansky and the Dvuroginsky suites of the Tunguska syncline and are dated as Lower Triassic (Induan stage).

4. Paleomagnetic study

One of us (J.D.-S.) had the opportunity to take samples along the Syradasay River section, although only one large oriented sample could be taken approximately every two flows from the lowermost lavas exposed in the top of the section. In total 31 samples were taken and their position in the section is indicated in Fig. 2. The samples were cut into several cubic (22 mm on a side) specimens.

4.1. Rock magnetism and demagnetisation of the NRM

The intensity of the natural remanent magnetisation (NRM) ranges from 0.2 to 8.0 A/m and the

susceptibility from 0.5 to $36 \cdot 10^{-3}$ SI. Isothermal remanent magnetisation (IRM) acquisition curves and decay curves were obtained for selected specimens for fields up to 1000 mT. Saturation was obtained with fields of between 70 and 500 mT. Backfield IRMs indicate an H_{CR} of 15–110 mT (Table 1).

Alternating and thermal demagnetisation of the NRMs show a similar range of properties. The median destructive fields (*mdf*) of the NRMs range from 10 to 100 mT. The unblocking temperatures are more constant, mainly at around 580°C or slightly above 620° or 650°C, but the median destructive temperature (*mdT*) of the NRM ranges widely, from 150° to 550°C. These experiments indicate that magnetite and haematite are the main carriers of the magnetisation, with a large range of grain sizes and coercivities. Titanomagnetites, demonstrated by low *mdT* values, may also be present.

Table 1
Comparison of the NRM and IRM magnetic properties

sample	NRM			IRM			
	mdf	Tb	mdT	H_{50}	H_{95}	H_{CR}	mdf
	mT	°C	mT	mT	mT	mT	mT
T10-4	20	585	500	30	100	30	15
T12-4	90	650	550	120	300	110	75
T19-4	>100	580	480	65	500	60	40
T22-4	50	620	530	60	200	40	25
T25-4	35	600	150	60	150	40	25
T27-6	90	620	250	120	200	110	75
T29-2	<10	580	200	20	90	15	<10
T30-4	20	580	380	20	65	15	<10

mdf = median alternating destructive field; T_b = NRM blocking temperature; *mdT* = temperature at which half of the NRM is destroyed; H_{50}, H_{95} = fields for which 50 and 95% of the saturation IRM is attained; H_{CR} = remanent coercive field

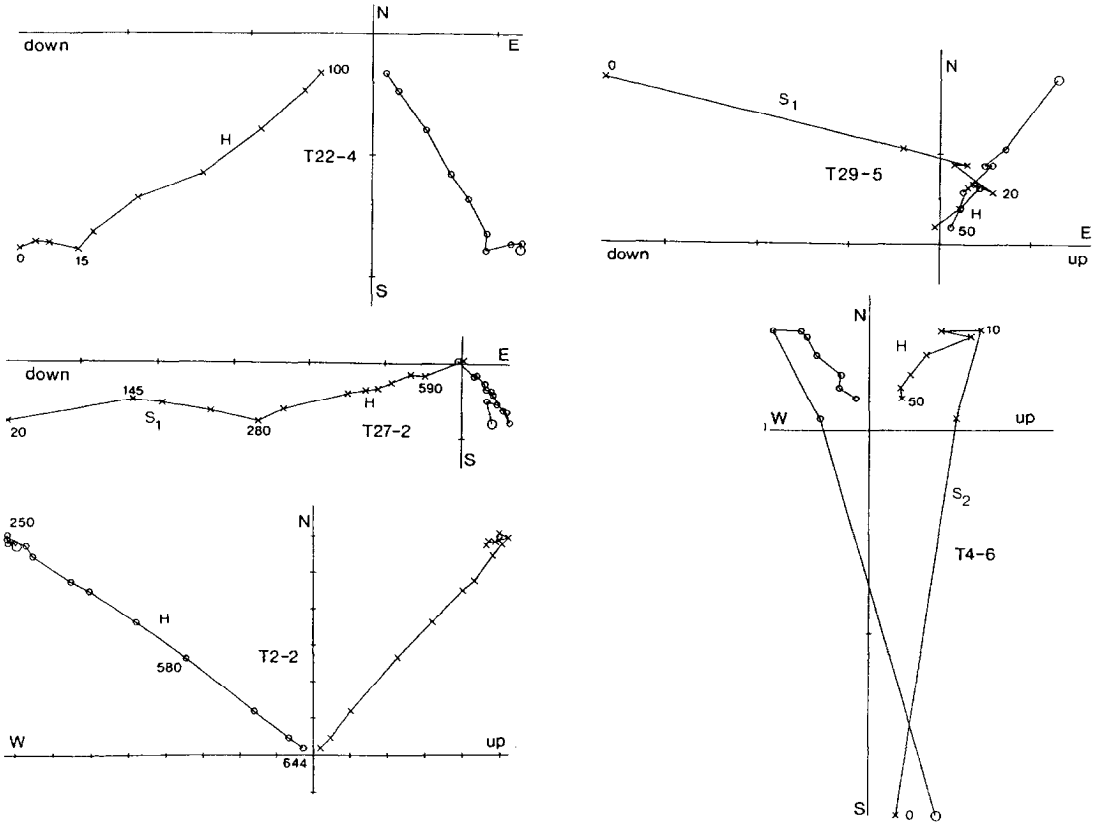


Fig. 3. Typical demagnetisation curves obtained with alternating field (samples T4-6, T22-4 and T29-5) and by heating (samples T2-2 and T27-2). O = horizontal plane; × = N–S vertical plane. Characteristic temperatures and field intensities are also given. All directions are in geographic coordinates.

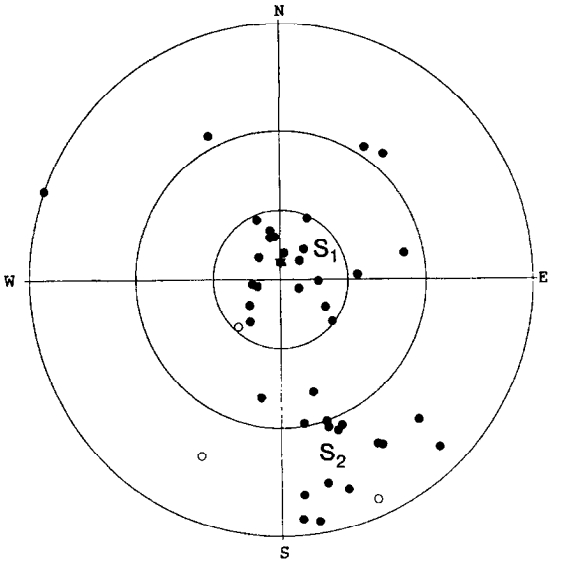


Fig. 4. Stereogram for component S in geographic coordinates for specimen directions. ● = positive inclinations; ○ = negative inclinations.

Even before any demagnetisation, the scatter of NRM directions from different specimens cut from the same sample is often very low. For more than half of the samples the κ-value of the Fisher statistics is higher than 500! At least one specimen per sample was stepwise thermally demagnetised and another using an alternating field up to 120 mT. Principal component analysis was used [25–27] and typical examples are given in Fig. 3. These examples often show an initial soft component removed by low fields (< 15 mT) or low temperatures (< 200°C). This soft component is called S. A second component is destroyed only with much higher fields and higher temperatures and is called H. Both demagnetisation methods give very similar results.

The stereogram of Fig. 4 shows that the S-directions may be subdivided into three groups. The first group (18 specimens), called S_1, is composed of directions that have positive inclinations above 60°. A second group (S_2, 17 specimens), is composed of directions with declinations to the south and positive inclinations below 45°. The third group (7 specimens) is composed of more scattered directions. Table 2 gives the mean directions of components S_1 and S_2 before and after tectonic correction.

If we apply the simple McElhinny fold test [28] to these two albeit somewhat arbitrarily separated groups that we call S_1 and S_2, we find that the κ-value for S_1 decreases during unfolding and that that for S_2 increases a little. It may be possible that component S_1, which is close to the present-day magnetic field, is a recent magnetisation caused by

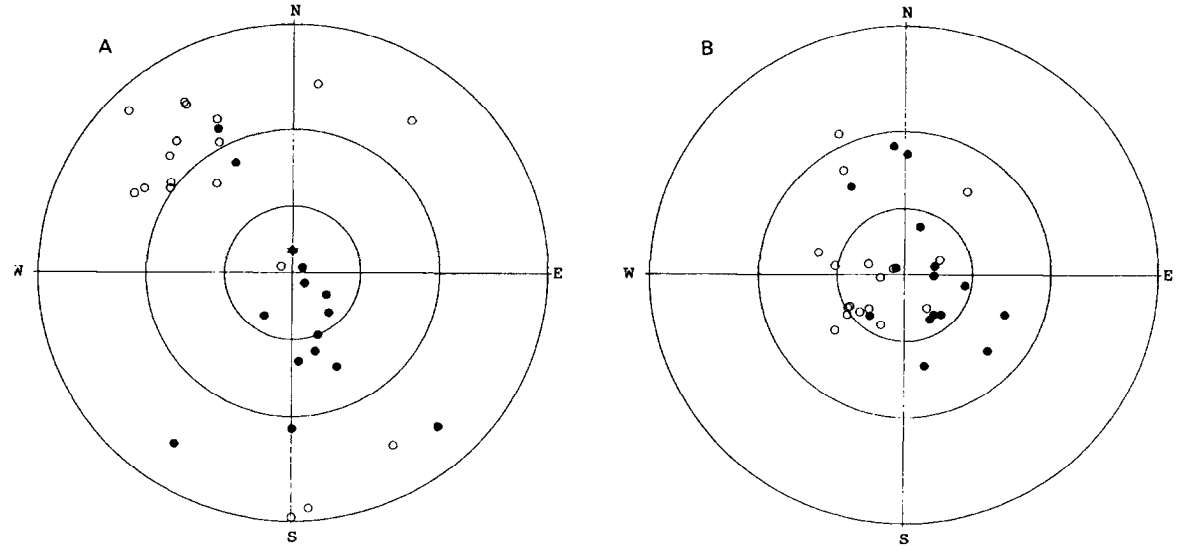


Fig. 5. Stereogram of sample means of component H. (A) *In-situ* position (geographic coordinates). (B) After tectonic correction (stratigraphic coordinates).

Table 2
Component *S*

Component	n/N	D	I	k	α_{95}
S1 in situ	18/13	28	+84	16.1	9
stratig	18/13	2	+41	4.8	16
S2 in situ	17/10	161	+19	18.4	8
stratig	17/10	130	+80	24.2	7
Present dipole field		0	+82		

n = number of specimens showing the component; N = corresponding number of samples; *in-situ* = mean direction in geographic coordinates; *stratig* = mean direction in stratigraphic coordinates. D, I = mean declination and inclination; κ = Fisherian κ-value; α_{95} = confidence angle

Table 3
Component *H*

Sample	mean directions retained					VGP	
Site	Dg	Ig	Ds	Is	level (m)	°N	°E
1	328	-12	240	-61	0	49	188
2	307	-28	205	-65	49	67	227
3	328	-13	278	-59	84	35	153
4	331	-28	147	-72	112	69	321
5	314	-22	231	-51	189	41	202
6	319	-20	264	-79	208	64	140
7	334	-21	288	-73	244	50	135
8	297	-20	226	-68	268	61	197
9	305	-29	235	-59	316	48	194
10	320	-40	230	-64	332	55	196
11	333	+37	335	-26	410	2	168
12	333	+24	330	-38	427	6	112
13	300	-22	240	-62	490	50	187
14	315	- 5	285	-51	550	26	150
15	7	-15	66	-73	630	49	38
16	150	-13	133	+42	690	12	127
17	137	+10	112	+44	742	18	146
18	180	-1	168	+49	794	13	95
19	180	+26	144	+68	847	36	111
20	176	-3	151	+67	959	34	106
21	214	+11	220	+66	1109	34	53
22	155	+45	100	+63	1167	39	149
23	164	+54	310	+85	1214	77	48
24	213	+67	329	+45	1283	41	301
25	60	+85	1	+38	1340	38	263
26	158	+60	17	+67	1405	66	237
27	131	+83	355	+35	1463	36	270
28	176	+51	73	+76	1518	63	156
(29	37	-15	36	-45	1568	12	51)
30	123	+72	91	+77	1658	60	141
(31	299	-84	299	-84	1713	65	109)
(31	138	+66	138	+66	1713	35	117)

D_g, I_g = mean direction in geographic coordinates; D_s, I_s = mean direction in stratigraphic coordinates; *level* = thickness above bottom of the section; *VGP* = position of virtual geomagnetic pole. Site position: 72.9°N, 84°E. Results marked with a parenthesis are not used in the means.

Table 4
Mean component *H* and fold test

Tauxe et al. statistics [31]

all directions	N	D	I	l_1	l_2	l_3
geographical coord.	29	150°	+28°	20.03	5.99	2.99
stratigraphical c.	28	84°	+74°	22.24	5.05	1.72
best unfolding ratio between 65 and 135 %.						

Fisher statistics	N	D	I	k	α_{95}
all directions(reversed directions inverted)					
geographical c.	29	152°	+28°	5.2	12°
stratigraphical c.	29	84°	+75°	7.1	10°
normal directions					
geographical c.	14	169°	+42°	4.6	19°
stratigraphical c.	14	84°	+75°	5.7	17°
reversed directions					
geographical c.	15	322°	-16°	10.0	12°
stratigraphical c.	15	264°	-71°	9.0	13°
Pole:	59°N	150°E	dp :17°	dm :18°	

N = number of results used in the statistics; D, I = mean declination and inclination; l_1, l_2, l_3 = eigenvalues in Tauxe et al. statistics; k = Fisherian concentration parameter; α_{95} = confidence angle.

magnetic viscosity or by recent weathering, and that component S_2 is an older one, perhaps pre-folding or syn-folding in age.

Fig. 5 and Table 3 show the mean component *H* for each sample. For many the directions of component *H* of different specimens cut from the same sample are very close together. The difference between specimens is less than 10° for 21 samples, and ranges between 10° and 20° for 8 samples. Two samples (29 and 31) show larger differences between specimens.

The different specimens of samples 29 and 31 show a soft normal component overprinting completely or partially a reversed component which has very similar coercivities and low unblocking temperatures, preventing a precise determination of the characteristic magnetization. These samples are considered as reversed, but their mean directions are not used in the final statistics.

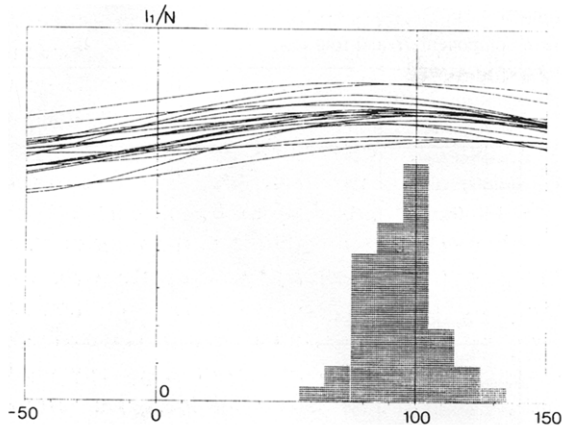


Fig. 6. Tauxe and Watson fold test [31]. Progressive unfolding of component H. Horizontal = unfolding ratio from −50% to 150%; vertical = major eigenvalue (I_1/N) ratio variations for 20 bootstrap draws. The histogram shows the best unfolding ratio for 100 bootstrap draws.

4.2. Fold tests

The results given in Table 4 show a large scatter with two antipodal directions. The fold test of Tauxe et al [29] and its further refinement by Tauxe and Watson [30] can be used with normal and reversed directions combined together without inverting a single polarity. The number of results is large enough to use a bootstrap method. Fig. 6 shows that the tilt-corrected directions are significantly better grouped than the *in-situ* directions. Complete (100%) unfolding provides the best statistical result.

The normal and reversed directions are antipodal; the angular distance between the normal and reversed axis is 4°, which is not significant. This places the reversal test in class A [31]. This reversal test indicates that there is no visible common overprint in the demagnetised directions. The positive fold test indicates that the magnetisation was acquired before the Late Triassic or Early Jurassic folding. Component H is then primary.

5. Magnetostratigraphy

5.1. Magnetic polarities found in the studied section

Both magnetic polarities are present in the Syradasay River section. The inclination of the mean

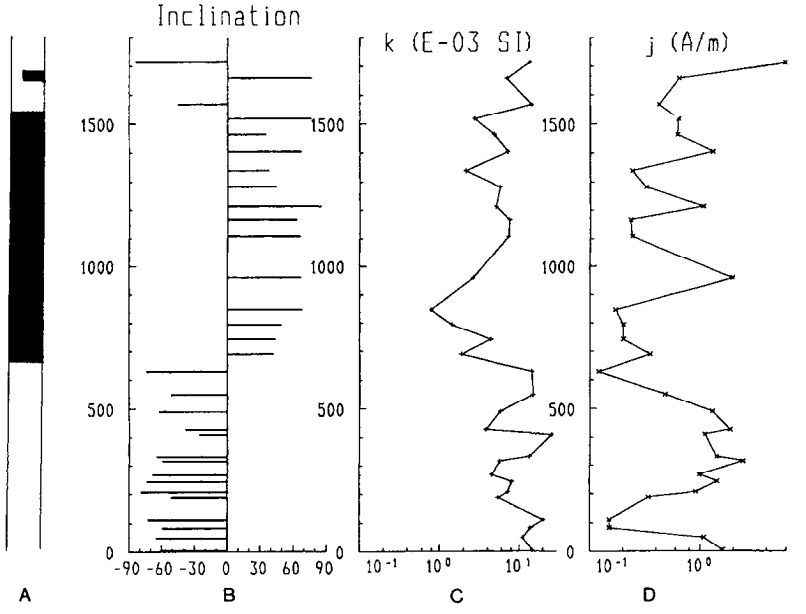


Fig. 7. Magnetostratigraphic interpretation of the Syradasay River section. (A) Polarity sequence. Black = normal; white = reversed. (B) Inclination in stratigraphic coordinates. (C) Susceptibility values in 10^{-3} SI,. (D) NRM intensity values in A/m. (Note log scale for C and D.)

direction is ±75°. With such high inclinations, the declination is of no use in the determination of the polarity, which is therefore given using the inclinations alone. The lower part, from level 1 up to level 15 (Fig. 2) is of reversed polarity and is 660 m thick. In the following part (900 m thick, from level 16 to 28) all the flows are normal. The polarity succession in the last part, which is 150 m thick, is more complex. Sample 30 looks normal, but samples 29 and 31 are reversed. We consider that this last part represents a reversed polarity zone with at least one short normal polarity event.

In conclusion the Syradasaysky suite is of reversed polarity, the Labaksky suite is normal, and the Verhnetamsky suite covers reversed and normal polarity. Fig. 7 shows the proposed polarity zonation of the Taimyr section.

5.2. *Synthesised polarity scale for the Siberian traps*

Several paleomagnetic and magnetostratigraphic studies have already been conducted by various authors for Siberia and Taimyr. However, unfortunately these results usually only ocuur in paleomagnetic catalogues [32–35], where detailed information on sampling, measurement and demagnetisation is not available. Moreover, many of the results were obtained at a time when demagnetisation was not carried out systematically, and therefore some results may be biased by undetected secondary components. The Siberian Platform traps are almost all flat-lying, which does not allow fold tests. However, many of these volcanics have a stable magnetisation and the polarity is usually well defined. The mean overall pole for the Siberian Platform lies at 53°N, 145°E.

With the above in mind, from the work of others giving a clear indication of which suites were sampled (and especially the stratigraphic position) we have carefully selected the results obtained after demagnetisation treatments. Normal polarity results represent more than 80% of all poles. The reversed polarities are obtained from the flows of the lower suites of the Noril'sk region, and in one case from the middle part of a section in Kotui Province. Reversed flows are also found in Taimyr at the bottom of several sections.

It is thus possible to construct a composite magnetostratigraphic section using both previous studies and the present results. Table 5 gives the typical sections, the terminology of the suites following Zolotukhin and Al'Mukhamedov [16], and the corresponding polarity. Compared to other results for Taimyr Province, our section is the first which allows direct lithostratigraphic correlation with other provinces. A composite magnetostratigraphic section is proposed in Fig. 8. This begins with a reversed

Table 5
Correlations between Siberian trap formations and magnetic polarity

Province	Noril'sk		Kureika		Kotui		W. Taimyr	
Cycle	Suite	Pol	Suite	Pol	Suite	Pol	Suite	Pol
V	Samoedsky	N						
	Kumginsky	N						
	Kharaelakhsky	N	Nerakarsky *	N	Meimechitic			
IV	Mokulaevsky *	N	Khona-makitsky	N	Delkansky		Verhnetamsky	NR
	Morongovsky *	N	Ayansky	N	Kogotsky	NR		
III	Nadezhdinsky	N	Dvuroginsky		Pravoboyarsky		Labaksky	N
	Tuklonsky	N	Tymersky		Arydzhangsky	N		
	Khakanchansky	N	Logachinsky					
II	Gudchikhinsky *	N						
	Syverminsky *	N						
I	Ivakinsky *	R					Syradasaysky	R

Pol = polarity; *N* = normal polarity alone; *R* = reversed polarity alone; *NR* = both polarities found. Correlation scheme after Zolotukhin and Al'Mukhamedov [16]. * Suites where reliable ages have been obtained (see text).

zone followed by a long normal one interrupted by some reversed flows in the Kogotsky suite of Kotui Province. This mixed part was not found (or is simply absent) in the Noril'sk region.

Geochemical correlations provide identical ages for Ivakinsky in Noril'sk and Syradasaysky in Taimyr, which are both reversed. The Labaksky suite is correlated with the suites forming Cycles II and III, which are normal everywhere. The Verhnetamsky suite is correlated with Cycle IV where the Kogotsky suite is of mixed polarity. More precisely, we propose to correlate the mixed zones of the Verhnetamsky suite found in Taimyr and the Kogotsky suite found in the Kotui region. The upper part of the traps (Cycle V) which is normal, is not represented in western Taimyr.

In summary, the synthesised scale structured around the results from western Taimyr gives magnetic polarities for a period corresponding to the entire volcanic episode. The bottom was emplaced during a reversed field and the top unambiguously belongs to a normal polarity. In the upper part of the lava sequence reversed directions of magnetisation are found in western Taimyr and in Kotui Province, in the Verhnetamsky and Kogotsky suites respectively (Table 5). These magnetisation directions can be interpreted as short reversed events that occurred during a predominantly normal period. The fact that this episode of alternating polarity is not seen in the Noril'sk and Kureika Provinces is possibly due to normal overprinting or to a local hiatus in the volcanism.

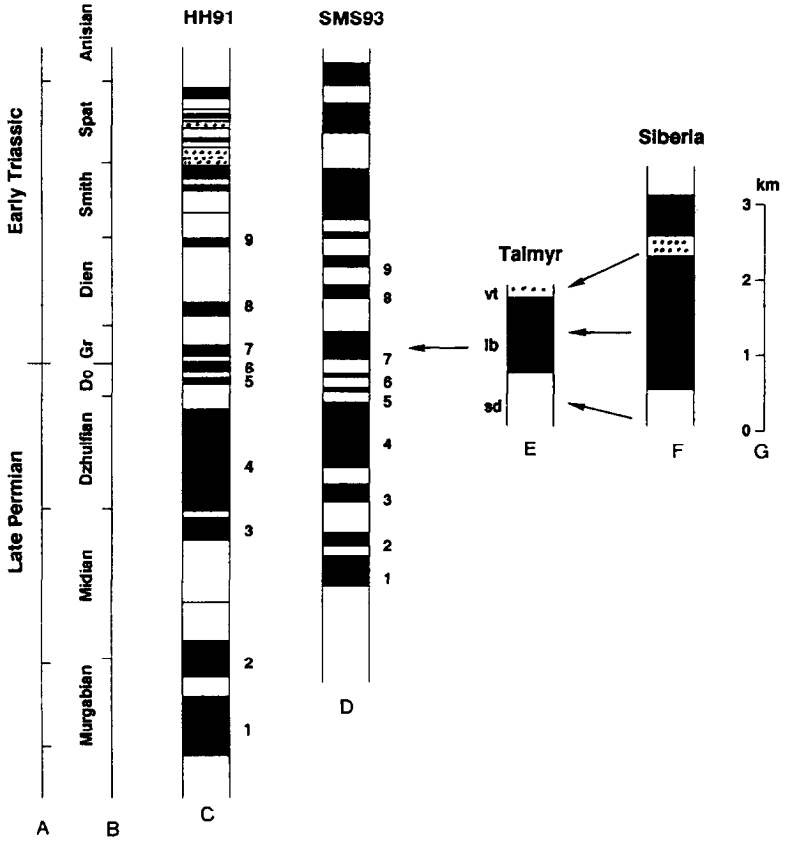


Fig. 8. Magnetostratigraphic timescales for the Late Permian and Early Triassic. HH91 = Haag and Heller [36] scale (note that the timescale is not linear); SMS93 = Steiner et al. [37] scale; Taimyr = present study; Siberia = composite scale for the Siberian traps. *Do* = Dorashamian; *Gr/Griesb* = Griesbachian; *Dien* = Deinerian; *Smit* = Smithian; *Spath* = Spathian. Dense black = normal; white = reversed; stipple = mixed or unknown polarity.

5.3. Magnetostratigraphic interpretation of the synthesised section

The succession of polarities of the Siberian and Taimyr traps, which corresponds to a very reduced number of polarity periods, may now be correlated to a standard scale covering the P/Tr boundary. For this purpose we shall use, as shown in Fig. 8, two composite magnetostratigraphic timescales for the Upper Permian and Lower Triassic—those of Haag and Heller [36] and Steiner et al. [37] (see also [38–41]). They look different, but they in fact show the same polarity sequences. The long reversed Kiaman superchron, which begins in the Upper Carboniferous, ends during the Tatarian stage (Upper Permian of the Russian Platform) and the Guadalupean (Upper Permian of North America) when the Illawara reversals begin. We have numbered the normal periods 1, 2 etc. in Fig. 8. The P/Tr boundary lies between No. 6 and 7. The differences between the two timescales is primarily due to different estimates of the duration of the Upper Permian stages and secondarily to different relative lengths of the normal and reversed polarity periods of the Lower Triassic. In both scales reversals are more frequent in the Lower Triassic (ca. two 2 or more per million years) than in the Upper Permian (less than one per million years).

The small number of reversals observed in the Siberian traps does not support long-lasting volcanism, and instead a much shorter period of volcanism is supported. The Siberian Platform composite section and the Taimyr section both show a reversed period at their bases which is always assigned to the Upper Permian. This reversed period is followed by one or at most two normal ones with an intermediate shorter reversed or mixed polarity period. These levels are assigned to the Lower Triassic. The paleontological ages given for the base of these traps (Upper Permian) and for their upper part (Lower Triassic) suggest a correlation of the normal lava sequence with normal period 7. This correlation is supported by reliable age determinations [11–13,18,20–24].

The question of the interpretation of the mixed Kogotski–Verhnetamsky cannot yet be answered. Either the mixed polarity is an undetected short reversed period within No. 7 or it corresponds to the reversed period between Nos. 7 and 8. In the first case, the total volcanic phase would have lasted less than 1 m.y., the maximum duration of normal period 7; in the second case it would have been longer, perhaps with a duration of 2 m.y. or more. Lind et al. (C.G. Czamanske, pers. commun.) analysed paleomagnetic results from the Noril'sk region and arrived at the same conclusion.

6. Conclusions

The Syradasay River section yelded a mean direction of $D = 84°$, $I = 75°$ with $\alpha_{95} = 10°$. The mean pole lies at 59°N, 150°E, with $d_p = 17°$ and $d_m = 18°$. Floral remnants from the base of this section are assigned to the Late Permian. The first 660 m of the section are of reversed polarity. The following 900 m are normal and the last 150 m are of mixed polarity. Corresponding floras for this last magneotozone are assigned to the Early Triassic. The magnetostratigraphic scale for this section is comparable with a composite scale for the Siberian Platform. The most reliable absolute age determinations from the Siberian Platform cover the range 248–250 Ma. Comparison with global composite magnetostratigrahic scales for the Late Permian and Early Triassic shows that the best correlation is with the first polarity zones of the Lower Triassic. The duration of the Siberian trap volcanism would then be between 0.5 and 2 m.y.

At a minimum the Siberian traps cover a surface of $1.5 \cdot 10^6$ km^2 with an average thickness of 3 km. The total volume can be estimated at $4.5 \cdot 10^6$ km^3, which erupted in $0.5–2 \cdot 10^6$ yr. The average output of lava was, then, between 2 and 9 km^3/yr. This may be compared with the Deccan trap production of 0.3–2.4 km^3/yr for extrusives and 2–8 km^3/yr for extrusives and intrusives combined [42] and with the average hotspot production since the Triassic (about 4.0 km^3/yr). The output of gases and dust expelled into the atmosphere is not known, but it must have been very high and continuous over very long periods. Moreover, we must bear in mind that the first cycles were much more explosive than the subsequent ones and that sulphides are present in large quantities (as ores, for instance) in the Noril'sk area. A high level of atmospheric SO_2 during certain periods and thus a catastrophic impact on the atmo-

sphere and climate is possible at the beginning of this volcanism. Our data are therefore consistent with volcanism the triggering mechanism of the P/Tr crisis.

Acknowledgements

We are indebted to M.B. Steiner, L. Tauxe, G. Czamanske, R. Montigny, H. Whitechurch, V. Courtillot and two anonymous reviewers for helpful comment and discussion. We are also very grateful to M.V. Verbitskaya, M.V. Turante and I.A. Dobruskina for determinations of the floral remnants. Visits to France by the Russian authors and to Russia by the French authors were paid for by the VNIGRI, by the Academy of Sciences of Russia, by the French CNRS (*Mission des Relations Internationales*) and by Programme DBT of the INSU. This is CNRS URA 323, CNRS URA 1316 and CNRS INSU contribution 746 (Programme DBT, *Dynamique globale*). [*FA*]

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